Every force has a source and a receiver

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Receiver / ES: objeto que recibe la fuerza / PT: objeto que recebe a força

The object a force acts on; also name its source.

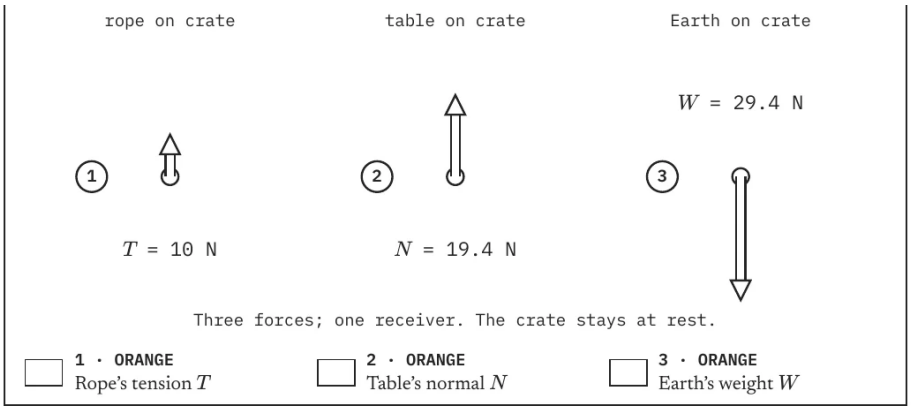
Also on your note page: Weight, Normal force, Tension.

B READ THE RELATIONSHIP

N + T − W = 0 for a crate at rest

As you read, find: source · receiver · contact.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg .

Given m = 3 kg T = 10 N up a y = 0

Rule W = mg . N + T − W = 0.

Work W = 3(9.8) = 29.4 N N = 29.4 − 10

Answer N = 19.4 N The rope shares the support.

E SAVE FOR THE FINAL MEETING

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

Read the physics: Receiver

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Ask: Which object receives the force? What other object pushes or pulls it? A rope pulls by contact. Gravity pulls even without contact.

02 HOW IT WORKS

Mass is measured in kilograms. Weight is a force in newtons. On Earth, multiply mass by gravitational field strength to find weight. Earth pulls downward; a table pushes perpendicular to its surface.

03 WHAT TO CHECK

A rope may carry part of a crate's weight. If the crate stays at rest, all upward forces together balance the downward force.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg . N = 19.4 N The rope shares the support.

Model check: N + T − W = 0 for a crate at rest.

05 CONNECT TO XJ

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

SAY IT BEFORE YOU WRITE

Who exerts the force, and on what object? The ___ exerts a force on the ___ toward ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Weight

ES: peso / PT: peso

Meaning in my words	How can you tell?
How would you explain weight to a friend? In my words, this means ___.	When a book rests on a table, which object supplies its weight? I can tell because ___.
Example or sketch	**Non-example and why**
Using 10 N/kg, what arrow represents the weight of a 3 kg book? My example shows ___.	A book slides down a ramp. Should its whole weight arrow point along the ramp? I would change ___ because ___.

Normal force

ES: fuerza normal / PT: força normal

Meaning in my words	How can you tell?
How would you explain normal force to a friend? In my words, this means ___.	A crate rests on a ramp. Which way does the ramp push the crate? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw the normal force on a book on a level table? My example shows ___.	Someone draws the normal force along the ramp. What would you rotate? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Tension

ES: tensión / PT: tração

Meaning in my words	How can you tell?
How would you explain tension to a friend? In my words, this means ___.	Which way does a slanted taut rope pull on a crate tied to its end? I can tell because ___.
Example or sketch	**Non-example and why**
Draw the force from a vertical cable on a hanging lamp. My example shows ___.	Could a loose rope push a cart forward when you move its free end toward the cart? I would change ___ because ___.

Receiver

ES: objeto que recibe la fuerza / PT: objeto que recebe a força

Meaning in my words	How can you tell?
How would you explain receiver to a friend? In my words, this means ___.	In “Earth pulls on the crate,” which object receives this force? I can tell because ___.
Example or sketch	**Non-example and why**
The rope pulls up on a lamp. On which object’s diagram do you draw that arrow? My example shows ___.	“The cart moves right.” Can you name a force receiver from that statement alone? I would change ___ because ___.

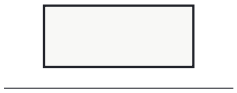
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

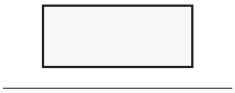
Draw a supported crate before and after an upward rope pull. Use the arrow lengths to show how the table's force changes.

Labels you can use: source · receiver · contact.

Upward support



Stronger rope pull



Tell a partner

The ___ exerts a force on the ___ toward ___.

On assignment: Receiver

SECTION E / BACK / FINAL MEETING ONLY Name _____

An XJ animation shows gravity from Earth and a forward arrow labeled "because it moves." The graphics desk needs approved force labels.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Simulation engineer**

Prepare the force layer for XJ. Name a source and receiver for each arrow you keep. Put motion on a separate layer.

☐ **Science graphics editor**

Sketch the graphic viewers will see. Have your producer challenge the source of each arrow before you approve its label.

Word bank: source · receiver · contact.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."

Editor's check: Identify another real source that would require its own force arrow.